Goal: Be able to understand the notation $\sum_{i=1}^{N} a_i$ and approximate the (signed) area beneath a curve.

Sigma (Summation) Notation

We want shorthand for sums with a predictable pattern. We use the Greek letter Sigma (Σ) to do so.

$$\sum_{i=m}^{n} a_i \stackrel{\text{def}}{=}$$

Example 1. Compute $\sum_{i=1}^{4} i^2$

Example 2. Compute $\sum_{j=0}^{6} (2^j + 1)$

Example 3. Rewrite $2^3 + 3^3 + 4^3 + \ldots + n^3$ in sigma notation.

Remark (Re-indexing)**.** You can start from any integer if

Example 4. Express $1^2 + 2^2 + (\sqrt{2}+1)^2 + (\sqrt{3}+1)^2 + \ldots + (\sqrt{n}+1)$ using sigma notation.

Theorem. Sums are linear in the sense that if k is a constant (i.e., not a function of the index), then:

(i) $\displaystyle\sum_{i=m}^{n} (a_i \pm b_i) = \sum_{i=m}^{n} a_i \pm \sum_{i=m}^{n} b_i$

(ii) $\displaystyle\sum_{i=m}^{n} k a_i = k \sum_{i=m}^{n} a_i$

Example 5. Compute $\displaystyle\sum_{i=1}^{n} k$ for any constant k.

Example 6. Show $\sum_{i=1}^{n} i = 1 + 2 + 3 + \ldots + n = \dfrac{n(n+1)}{2}$

The next theorem is not required, but it is beneficial to know since it can speed up calculations.

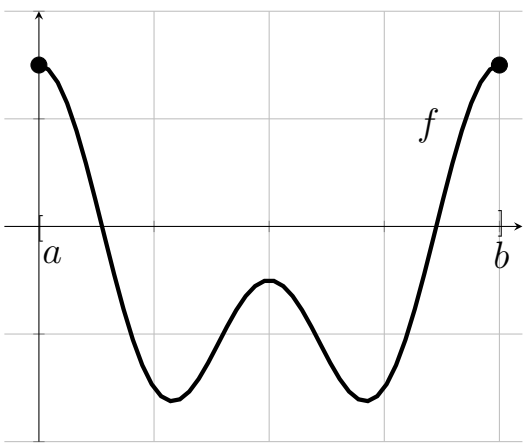
Theorem. (i) $\sum_{i=1}^{n} k = kn$ for constant k

(ii) $\sum_{i=1}^{n} i = \dfrac{n(n+1)}{2}$

(iii) $\sum_{i=1}^{n} i^2 = \dfrac{n(n+1)(2n+1)}{6}$

(iv) $\sum_{i=1}^{n} i^3 = \left[\dfrac{n(n+1)}{2}\right]^2$

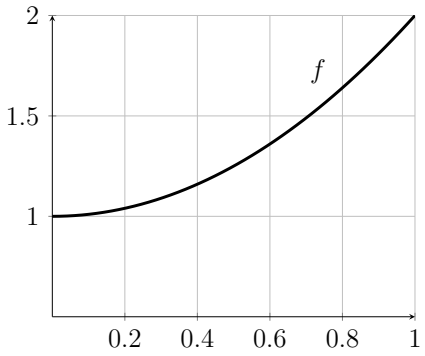
The Area Problem



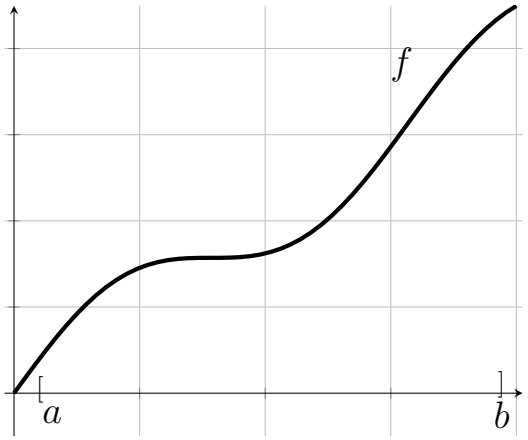
Problem: Given a function f defined on a closed interval $[a, b]$, compute the (signed) area between the graph of f and the x-axis.

"Solution:"

Example 7. Approximate the area of the region beneath the graph of $f(x) = x^2 + 1$ on $[0, 1]$ using 5 rectangles.



This is an example of a **left** _____ ("left" is referring to how we chose the _____). How can we do this in general?



1. Divide $[a, b]$ into N equal regions.
2. Determine the dimensions of each rectangle.

Definition. The **left Riemann sum** of f over $[a, b]$ is defined as:

$$L_N \overset{\text{def}}{=}$$

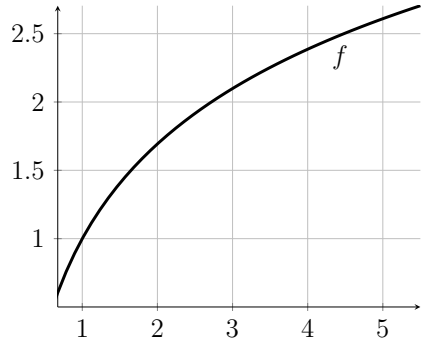
Similarly, the **right Riemann sum** is defined as:

$$R_N \overset{\text{def}}{=}$$

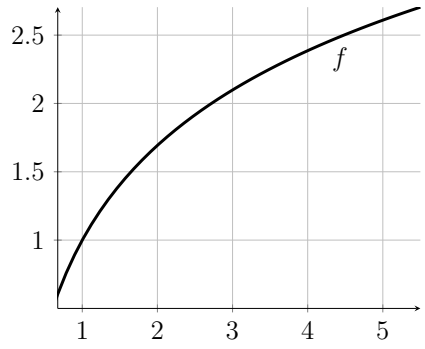
where $\Delta x =$

Example 8. Consider $f(x) = \ln x + 1$ on the interval $[1, 4]$:

(a) Compute R_6 [i.e., the right Riemann sum with 6 rectangles].



(b) Repeat for L_6



(c) Set up (but do not compute) L_N and R_N.